P12928 · KPYR_RAT

Protein[i]
Pyruvate kinase PKLR

Gene[i]
Pklr

Status[i]
UniProtKB reviewed (Swiss-Prot)

Organism[i]
Rattus norvegicus (Rat)

Amino acids
574 (go to sequence)

Protein existence[i]
Evidence at protein level

Annotation score[i]
5/5

Function[i]

Pyruvate kinase that catalyzes the conversion of phosphoenolpyruvate to pyruvate with the synthesis of ATP, and which plays a key role in glycolysis. Also produces the side product 2-phospholactate which can inhibit fructose-2,6-bisphosphate production. 2-phospholactate can be dephosphorylated by PGP which prevents the inhibition of fructose-2,6-bisphosphate production and allows glycolysis to occur. By Similarity

Miscellaneous

There are 4 isozymes of pyruvate kinase in mammals: L, R, M1 and M2. L type is major isozyme in the liver, R is found in red cells, M1 is the main form in muscle, heart and brain, and M2 is found in early fetal tissues. Curated

Catalytic activity[i]

Rhea:18157

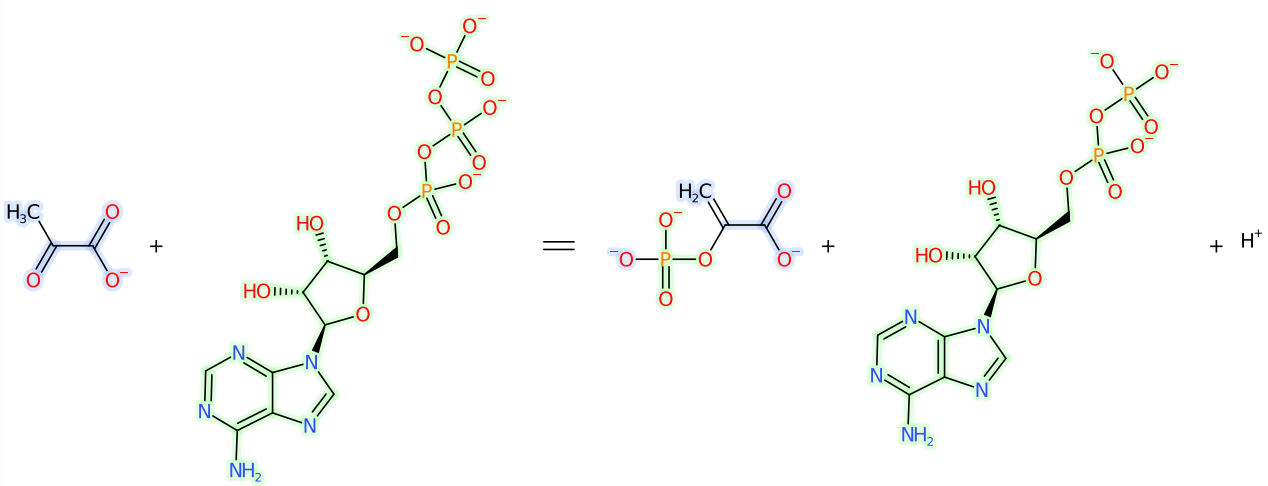
pyruvate + ATP = phosphoenolpyruvate + ADP + H^+ By Similarity
This reaction proceeds in the backward direction. By Similarity
EC:2.7.1.40 (**UniProtKB** | **ENZYME** | **Rhea**)

Hide reaction details

Reaction | Atom map

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Color atoms ☑, Number atoms ☐, Confidence = 0.34795



pyruvate » ATP » phosphoenolpyruvate » ADP » H^+ »

Rhea:86567

(S)-lactate + ATP = (2S)-2-phospholactate + ADP + H^+ By Similarity

View reaction details

Cofactor[i]

Protein has several cofactor binding sites:

Mg^{2+} (UniProtKB | Rhea | CHEBI:18420) By Similarity
Mn^{2+} (UniProtKB | Rhea | CHEBI:29035) By Similarity

K^+ (UniProtKB | Rhea | CHEBI:29103) By Similarity

Activity regulation[i]

Allosterically activated by fructose 1,6-bisphosphate. By Similarity

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Features

Showing features for binding site[i], site[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Binding site		116	substrate By Similarity	
+	Binding site		118	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		118-121	ATP (UniProtKB | ChEBI) By Similarity	Tools Add
+	Binding site		120	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		156	K^+ (UniProtKB | ChEBI) By Similarity	
+	Binding site		157	K^+ (UniProtKB | ChEBI) By Similarity	

Expand table

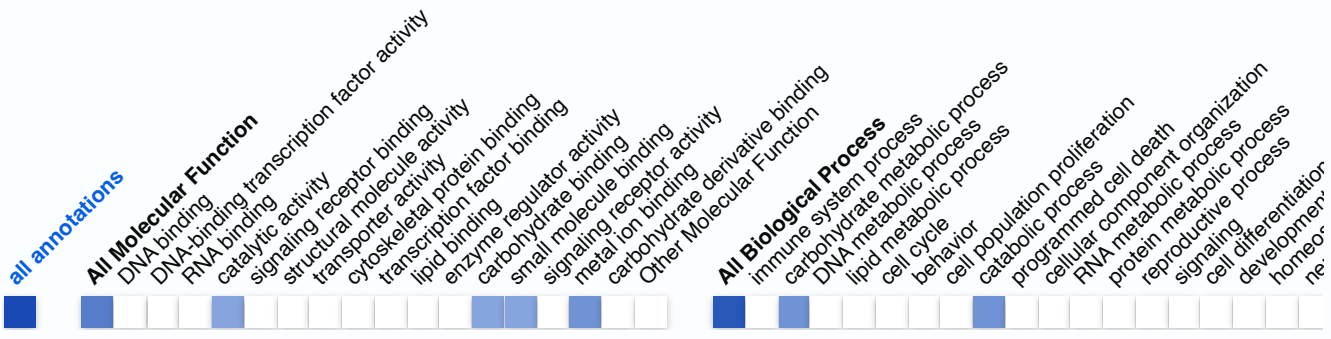
Gene Ontology[i]

GO annotations | GO-CAM models New

Gene Ontology (GO) annotations organized by slimming set.

Slimming set:

Animals and fungi (Opisthokonta)



Cell color indicative of number of GO terms

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ASPECT	TERM
Molecular Function	monosaccharide binding Source:RGD 1 Publication
Molecular Function	potassium ion binding Source:InterPro
Molecular Function	pyruvate kinase activity Source:RGD 5 Publications
Biological Process	cellular response to epinephrine stimulus Source:RGD 1 Publication
Biological Process	cellular response to insulin stimulus Source:RGD 1 Publication
Biological Process	glycolytic process Source:RGD 2 Publications
Biological Process	pyruvate biosynthetic process Source:RGD 2 Publications
Biological Process	response to ATP Source:RGD 1 Publication
Biological Process	response to cAMP Source:RGD 1 Publication

Expand table

Access the complete set of GO annotations on QuickGO

Keywords[i]

Molecular function

#Allosteric enzyme
#Kinase
#Transferase

Biological process

#Glycolysis

Ligand

#ATP-binding
#Magnesium
#Manganese
#Metal-binding
#Nucleotide-binding

More keywords

Enzyme and pathway databases

Reactome

R-RNO-70171 Glycolysis
R-RNO-70268 Pyruvate metabolism

SABIO-RK

P12928

UniPathway

UPA00109UER00188

ENZYME

Search…

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Protein names

Recommended name

Pyruvate kinase PKLR

EC number

EC:2.7.1.40 (**UniProtKB** | **ENZYME** | **Rhea**) By Similarity

Alternative names

L-PK

Pyruvate kinase isozymes L/R

Gene names

Name

Pklr

Organism names

Taxonomic identifier

10116 (**NCBI**)

Organism

Rattus norvegicus (Rat)

Taxonomic lineage

Eukaryota > Metazoa > Chordata > Craniata > Vertebrata > Euteleostomi > Mammalia > Eutheria > Euarchontoglires > Glires > Rodentia > Myomorpha > Muroidea > Muridae > Murinae > Rattus

Accessions

Primary accession

P12928

Secondary accessions

P04763

Q64618

Proteomes

Identifier

UP000002494

Component

Chromosome 2

Organism-specific databases

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AGR
RGD:3336

RGD
3336 Pklr

Subcellular Location

UniProt Annotation | GO Annotation

cytoplasm Source:RGD
cytosol Source:RGD
plasma membrane Source:RGD

Complete GO annotation on QuickGO

Phenotypes & Variants

Chemistry

ChEMBL
CHEMBL3089



Expression

Gene expression databases

Bgee
ENSRNOG00000020420 Expressed in duodenum and 14 other cell types or tissues

ExpressionAtlas
P12928 baseline and differential

Interaction

Subunit

Homotetramer By Similarity

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BioGRID
246786 1 interactor

FunCoup
P12928 317 interactors

IntAct
P12928 1 interactor

MINT
P12928

STRING
10116.ENSRNOP00000027700

Chemistry

BindingDB
P12928

Structure[i]



3D structure databases

AlphaFoldDB
P12928

ModBase
Search…

SMR
P12928

PDBe-KB
Search…

Family & Domains[i]

Sequence similarities[i]

Belongs to the **pyruvate kinase family**. Curated

Family and domain databases

- View the Phylogenomic databases for this entry within the **Homologs section**.
- View all family and domain features for this entry's canonical sequence in the **UniParc Feature Viewer**.

CDD
cd00288 Pyruvate_Kinase 1 hit

FunFam
2.40.33.10:FF:000001 Pyruvate kinase 1 hit

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Gene3D

3.20.20.60 Phosphoenolpyruvate-binding domains 1 hit

2.40.33.10 PK beta-barrel domain-like 1 hit

3.40.1380.20 Pyruvate kinase, C-terminal domain 1 hit

InterPro

View protein in InterPro

IPR001697 Pyr_Knase

IPR015813 Pyrv/PenolPyrv_kinase-like_dom

IPR040442 Pyrv_kinase-like_dom_sf

IPR011037 Pyrv_Knase-like_insert_dom_sf

More InterPro links

NCBIfam

NF004491 PRK05826.1 1 hit

NF004978 PRK06354.1 1 hit

TIGR01064 pyruv_kin 1 hit

PANTHER

PTHR11817 PYRUVATE KINASE 1 hit

PRINTS

PR01050 PYRUVTKNASE

PROSITE

View protein in PROSITE

PS00110 PYRUVATE_KINASE 1 hit

Pfam

View protein in Pfam

PF00224 PK 1 hit

PF02887 PK_C 1 hit

SUPFAM

SSF51621 Phosphoenolpyruvate/pyruvate domain 1 hit

SSF50800 PK beta-barrel domain-like 1 hit

SSF52935 PK C-terminal domain-like 1 hit

MobiDB

Search…

Sequence & Isoform[i]

Align isoforms (2) **Add isoforms**

Sequence status[i]

Complete

This entry describes **2** isoforms[i] produced by **Alternative splicing**.

P12928-1

This isoform has been chosen as the **canonical** sequence. All positional information in this entry refers to it. This is also the sequence that appears in the downloadable versions of the entry.

Name

R-type

Synonyms

PKR

See also

sequence in **UniParc** or sequence clusters in **UniRef**

Length

574

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Last updated
1995-02-01 v2

MD5 Checksum[i]
E310DE6CCC4190DEB38EF2EC34F65859

MSVQENTLPQQLWPWIFRSQKDLAKSALSGAPGGPAGYLRRASVAQLTQELGTAFFQQQQLPAAMADTFLE
HLCLLDIDSQPVAARSTSIIATIGPASRSVDRLKEMIKAGMNIARLNFSHGSHEYHAESIANIREATESFATSPLSY
RPVAIALDTKGPEIRTGVLQGGPESEVEIVKGSQVLVTVDPKFQTRGDAKTVWVDYHNITRVVAVGGRIYIDDG
LISLVVQKIGPEGLVTEVEHGGILGSRKGVNLPNTEVDLPGLSEQDLLDLRFGVQHNVDIIFASFVRKASDVLAV
RDALGPEGQNIKIISKIENHEGVKKFDEILEVSDGIMVARGDLGIEIPAEKVFLAQKMMIGRCNLAGKPVVCATQ
MLESMITKARPTRAETSDVANAVLDGADCIMLSGETAKGSFPVEAVMMQHAIAREAEAAVYHRQLFEELRRAA
PLSRDPTEVTAIGAVEASFKCCAAAIIVLTKTGRSAQLLSQYRPRAAVIAVTRSAQAARQVHLSRGVFPLLYREP
PEAIWADDVDRRVQFGIESGKLRGFLRVGDLVIVVTGWRPGSGYTNIMRVLSVS

P12928-2

Name
L-type
Synonyms
PKL

See also
sequence in **UniParc** or sequence clusters in **UniRef**
Differences from canonical
1-33: MSVQENTLPQQLWPWIFRSQKDLAKSALSGAPG → ME Curated

Features

Showing features for alternative sequence[i], sequence conflict[i].

±	TYPE	ID	POSITION(S)	DESCRIPTION	
+	Alternative sequence	VSP_002884	1-33	in isoform **L-type** Curated	Tools Add
+	Sequence conflict		130	in Ref. 1; AAA41882/AAA41883 Curated	
+	Sequence conflict		322	in Ref. 1; AAA41882/AAA41883 Curated	
+	Sequence conflict		498	in Ref. 3; AAA41880 Curated	
+	Sequence conflict		501	in Ref. 3; AAA41880 Curated	

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Coding sequence diversity
#Alternative splicing

Technical term
#3D-structure
#Reference proteome

Sequence databases

PIR
A27427 KIRTPR
A92940 KIRTPL

RefSeq
NP_036756.3 NM_012624.3 [P12928-2]
XP_006232655.1 XM_006232593.5 [P12928-1]

NUCLEOTIDE SEQUENCE	PROTEIN SEQUENCE	MOLECULE TYPE	STATUS
M17091 EMBL · GenBank · DDBJ	AAA41882.1 EMBL · GenBank · DDBJ	Genomic DNA	
M17088 EMBL · GenBank · DDBJ	AAA41882.1 EMBL · GenBank · DDBJ	Genomic DNA	
M17089 EMBL · GenBank · DDBJ	AAA41882.1 EMBL · GenBank · DDBJ	Genomic DNA	
M17090 EMBL · GenBank · DDBJ	AAA41882.1 EMBL · GenBank · DDBJ	Genomic DNA	
M17091	AAA41883.1		

Expand table

Genome annotation databases

Ensembl
ENSRNOT00000027700.8
ENSRNOP00000027700.3
ENSRNOG00000020420.8 [P12928-1]
ENSRNOT00000103387.2
ENSRNOP00000093132.1
ENSRNOG00000020420.8 [P12928-2]
ENSRNOT00065042893 ENSRNOP00065035101
ENSRNOG00065024916 [P12928-1]

GeneID
24651

KEGG
rno:24651

UCSC
RGD:3336 rat [P12928-1]

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Similar Proteins[i]

UniRef clusters[i]



Homologs[i]

Orthologs & paralogs[i]



Phylogenomic databases

GeneTree
ENSGT00390000008859

HOGENOM
CLU_015439_0_1_1

InParanoid
P12928

OMA
HQGRYDR

OrthoDB
44365at9989

PhylomeDB
P12928

eggNOG
KOG2323 Eukaryota

Organism-specific databases

AGR
RGD:3336

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